NETSOL | PROJECT REVIEW

Welcome to our research demonstration

Explainable Multimodal AI for Behavioral Analysis

AV-ASD + ASD-FEAT | IMPLEMENTATION BRIEF

What we built. What we used. Where the data came from.
Validated workflows, recorded results and next steps.

Presented by Muhammad Danial

Team: Muhammad Danial | Hamza Rasheed | Muhammad Freeh

12 October 2026 | Prepared for NETSOL review

Independent team project. Prepared for review; no NETSOL endorsement implied.

01 / PURPOSE AND COMPLETED WORK

What the project does

Simple explanation: Upload a short video. The AV-ASD model calculates audio and visual features, predicts behavioral-label scores, explains which features influenced them, and produces a downloadable report.

A separate ASD-FEAT research track compares prepared audio, RGB and human-coded behavioral features. Its results answer a different question and are shown separately.

Completed component	What we built
Data preparation	Checked annotations, clip availability, feature values, IDs and split overlap.
Model experiments	Compared audio, visual and combined baselines, early fusion and late fusion.
Evaluation	Saved frozen test results; completed grouped cross-validation and error analysis.
Model inspection	Built linear feature/branch explanations, retrained ablations and simulated missing-branch checks.
Reproducibility	Exported standalone scripts and verified Windows training/inference reproduction.
Streamlit interface	Results pages, compatible feature-CSV analysis, local MP4 upload, summary, explanations and HTML/JSON reports.

Current scope

This is an explainable behavioral research prototype. It does not confirm autism, determine a clinical stage, recommend treatment or provide a validated percentage chance of autism.

02 / DATASET PROVENANCE

Where the data came from

AV-ASD: labeled video clips [1, 2]

Dataset annotations and source URLs came from the authors' AV-ASD repository. The repository lists original YouTube and Facebook videos; this project downloaded available source media and extracted the annotated clips. No new child dataset was collected by our team.

Local project audit	Count
Candidate labeled clips	928
Usable clips	527
Training / validation / test	312 / 108 / 107
Unavailable source clips / duration failures	399 / 2
Behavioral labels	10

Example labels include repetitive upper-limb movements, spinning, eye-contact avoidance and verbal non-response. Predictions are not confirmed observations. Source-video separation passed, but it does not prove separation of children.

ASD-FEAT: released derived features [3, 4]

Prepared feature archives and metadata were obtained from the ASD-FEAT Zenodo release. The authors describe a derived-feature release, rather than raw audio/video. We used released audio summaries, I3D RGB representations and human-coded behavior measurements.

Aligned E6 data	Count
Development recordings / participants	1,510 / 259
Official test recordings / participants	162 / 30
Audio / RGB / behavior inputs	224 / 1,024 / 12
Positive test recordings / positive participants	12 / 1

Dataset reuse permissions and release-specific terms require review before broader publication or redistribution. These data tracks must not be combined into one accuracy claim.

03 / TECHNOLOGY AND METHODS

What we used, and why

Tool or method	Role in our implementation
Python, NumPy, pandas	Scripts, numerical calculations, feature tables and alignment checks.
librosa + SoundFile	Audio loading; MFCC, energy and zero-crossing summaries.
OpenCV + FFmpeg	Decode clips, extract audio, sample frames and retry difficult video decoding.
scikit-learn	Logistic Regression, Random Forest, StandardScaler, PCA and evaluation metrics.
joblib	Save and reload trusted local fitted models and schema bundles.
Streamlit	Interactive results dashboard, uploads and downloadable reports.
PowerShell + Git/GitHub	Windows execution, version control and reviewable project updates.
MediaPipe - exploratory pilots	Pose quality and image-embedding feasibility checks; not the final ASD-FEAT raw-video extractor.

How the models were built

AV-ASD: 44 audio features + 14 handcrafted visual features. StandardScaler and one-vs-rest balanced Logistic Regression form the primary model. Random Forest and late fusion were compared. The validation-selected threshold is 0.30.

ASD-FEAT: Five participant-disjoint development folds. Scaling and PCA were fitted within each fold. RGB and audio branches used 32 PCA components; Logistic Regression classified the processed features at threshold 0.50.

RGB plus human-coded behavior was selected after exploratory development comparisons. Its compatible input has 1,036 values. It uses released I3D features; it is not the same pipeline as the 58-feature MP4 page.

04 / EVIDENCE AND VALIDATION

What the results show

Experiment	Saved result	Meaning
AV-ASD combined LR	Macro F1: 0.2937 Micro F1: 0.3544	107 test clips; behavior-label prediction remains limited.
ASD-FEAT RGB + behavior	Mean development ASD F1: 0.3142	Participant-grouped CV; exploratory model selection.
ASD-FEAT frozen test	ASD F1: 0.3571 Precision: 0.2273 Recall: 0.8333	162 recordings; TP 10, FP 34, FN 2, TN 116.

Important interpretation: All 12 positive ASD-FEAT test recordings belong to one participant. This test cannot establish performance across different positive children. Scores and metrics from the two tasks are not directly interchangeable.

Completed application checks

Validation	Observed outcome
Prepared CSV input handling	10/10 accepted/rejected-case checks passed.
MP4 feature reproduction	5/5 existing clips matched saved features, probabilities and predictions.
MP4 edge cases	5/5 passed: silent and short clips accepted; no-audio, long and corrupt files rejected.
Explanation reconstruction	107 recordings x 10 labels = 1,070 checks passed; maximum score difference 6.66e-16.
User-interface behavior	Changing the video cleared the previous report; exported summary and explanations were inspected.

These checks verify the tested software behavior and model arithmetic. They do not establish accuracy on arbitrary uploaded videos or clinical reliability.

05 / DEMO AND NEXT STEPS

How to present the work

A short demonstration

1. Results: Show the baseline and grouped-CV comparisons.

2. Model inspection: Explain fusion, branch ablation and feature contributions.

3. Video upload: Open Video Behavior Analysis and upload a permitted MP4.

4. Analyze: Show the ten label scores, fixed threshold, input quality and readable summary.

5. Explain and export: Expand a label's top contributions; download HTML or JSON.

The MP4 page supports an audio track, at most 120 seconds, 7,200 frames and 200 MiB. Silent audio is accepted and flagged; a missing audio track is rejected.

Remaining work and planned extensions

Submission updates: Add the validated video-upload extension to the final report, demo guide and README; then prepare the GitHub update.

Research extensions: Evaluate more diverse recordings and positive participants; investigate behavior timing within videos and stronger temporal models. These are planned, not completed capabilities.

Deployment review: Confirm dataset permissions, privacy requirements, feature provenance and independent performance before wider use.

Dataset sources

1. AV-ASD authors' repository
https://github.com/ShijianDeng/AV-ASD

2. AV-ASD paper: Hear Me, See Me, Understand Me
https://arxiv.org/abs/2406.02554

3. ASD-FEAT authors' code and dataset description
https://github.com/sidrah-liaqat/asd-feat

4. ASD-FEAT dataset DOI
https://doi.org/10.5281/zenodo.22261227

Evidence basis: supplied project results, Windows verification output and inspected exported reports. Dataset provenance checked against authors' repositories on 12 October 2026; the Zenodo landing page could not be retrieved during this check. This brief does not rerun training.